AutoShotV2: Calibrated Classification and Post-Processing for Short-Video Shot Boundary Detection

Senh Manh Tang, Tien Manh Do, and Thai Son Tran

Abstract—Shot boundary detection (SBD) is a fundamental preprocessing step for video indexing, retrieval, summarization, and automated editing. However, modern short-video platforms create a difficult operating regime: videos are brief, shots are extremely short, transitions are visually complex, and boundary frames are rare compared with non-boundary frames. This paper presents AutoShotV2, a lightweight improvement over the AutoShot architecture for short-video SBD. Instead of re-searching or retraining the full backbone, AutoShotV2 freezes the AutoShot feature extractor and trains a new two-branch classification head over 4864-dimensional cached features. The head is optimized with Focal Loss and complementary one-hot and boundary-aware many-hot supervision. At inference time, the primary boundary logits are calibrated by temperature scaling and stabilized by one-dimensional Gaussian smoothing before thresholding. On the SHOT test set of 200 short videos, AutoShotV2 reaches an F1-score of 0.8545, with precision of 0.8557 and recall of 0.8533, improving over the reproduced AutoShot baseline by 1.40 F1 points and over the reported TransNetV2 baseline by 5.52 points. Experiments on BBC and ClipShots show that the same fixed deployment configuration transfers well to documentary footage but remains limited on open-world gradual transitions. The code is available at https://github.com/DoTienManhCoder/AutoShotV2.

Index Terms—Shot boundary detection, short video, AutoShot, TransNetV2, Focal Loss, temperature scaling, Gaussian smoothing, video understanding.

I. Introduction

Effectively parsing and understanding video content is a fundamental requirement for many downstream applications, such as video indexing, content retrieval, summarization, and automated editing. Within this pipeline, shot boundary detection (SBD) serves as a critical preliminary step. Its primary objective is to partition a video into temporally coherent shots by identifying the exact frames where camera views, scene contents, or editing states transition [1], [2]. While classical SBD methods rely on hand-crafted cues (e.g., color histograms and motion differences), these approaches have proven fragile under complex scenarios involving fast camera motion, flashes, and modern editing styles.

For SBD research, deep learning models have significantly advanced the state-of-the-art. Architectures like TransNet and TransNetV2 [5], [6] successfully model spatial-temporal representations directly from frame sequences. Concurrently, the rise of short-video platforms (e.g., TikTok, YouTube Shorts) has introduced a new, challenging operating regime [14]. To address this, AutoShot [4] targeted short-video SBD by leveraging neural architecture search (NAS) to find an optimal 3-D convolutional backbone, demonstrating strong performance on the newly introduced SHOT dataset. However, further improving these models encounters a severe bottleneck: the NAS pipeline requires immense computational overhead for supernet training and architecture evaluation. Furthermore, short-video SBD suffers from extreme class imbalance (boundary frames are exceptionally rare) and complex visual effects that act as hard negatives.

Following the promising strategy of separating representation learning from task-specific adaptation, we aim to study a more efficient scaling and refinement property for short-video SBD. Instead of re-running the expensive NAS process, we present a simple yet effective method to scale the performance of the existing architecture, termed AutoShotV2. Our core technical improvement is the introduction of a dual-branch classification head applied over the frozen 4864-dimensional features extracted by the AutoShot backbone. To handle the extreme class imbalance, we optimize this head using Focal Loss [11], focusing the learning capacity on hard negatives and ambiguous transitions. Furthermore, we devise a boundary-aware many-hot supervision strategy in the auxiliary branch to provide denser temporal gradients without compromising the precise one-hot localization of the primary branch.

Finally, adapting deep models to threshold-based decision tasks often suffers from overconfidence and local noise. To mitigate this, we introduce a robust post-processing paradigm. At inference time, the primary boundary logits are calibrated using temperature scaling [12] to adjust the numerical scale of the predictions without altering their ranking. Subsequently, a one-dimensional Gaussian temporal filter is applied to smooth out isolated, false-positive peaks caused by sudden flashes or camera shakes.

Based on these core designs—frozen robust representations, dual-branch focal supervision, and calibrated temporal smoothing—we develop AutoShotV2. On the SHOT test set, AutoShotV2 achieves an F1-score of 0.8545, the best result among the evaluated methods, improving over the reproduced AutoShot baseline and the reported TransNetV2 baseline by 1.40 and 5.52 F1 points, respectively.

In summary, our main contributions are:

- We propose AutoShotV2, an efficient and highly accurate SBD pipeline that significantly improves short-

video boundary detection without the computational burden of retraining the NAS-searched backbone.
- We design a novel two-branch classification head trained with Focal Loss and boundary-aware many-hot supervision to address the extreme class imbalance inherent in short videos.
- We introduce a calibrated inference pipeline leveraging temperature scaling and Gaussian smoothing to ensure stable, threshold-based boundary decisions.
- We provide extensive experiments on SHOT, BBC, and ClipShots, demonstrating strong performance gains and identifying important cross-domain limitations for future research.

II. Related Work

Shot boundary detection. Shot boundary detection has long been an active research area in video processing. Early systems primarily measured frame-to-frame discontinuities using hand-crafted visual features. Methods relying on histogram differences, edge-change ratios, motion compensation, and thresholding-based heuristics are computationally simple. However, they are prone to mistaking lighting changes, camera motion, or object movement for actual shot boundaries [3]. Comprehensive survey studies categorize the major challenges in this domain into abrupt cuts, gradual transitions, illumination changes, object/camera motion, and dataset-dependent evaluation protocols [1], [2].

Deep SBD models. Following the success of deep learning in vision tasks, deep models were introduced to learn temporal patterns that are difficult to encode manually [18], [21]. TransNet models SBD as a dense frame-level prediction task over a sequence of low-resolution frames [6]. TransNetV2 further improves this concept by incorporating dilated 3-D convolutions, frame-similarity features, color histograms, and dual prediction heads for both single-frame and all-transition labels [5]. These models offer a fast and accurate inference pipeline, establishing them as strong baselines for practical SBD applications.

AutoShot and short-video SBD. Most existing datasets, such as BBC, RAI, and ClipShots, mainly represent documentaries, broadcast content, or longer user-generated videos. Recognizing this gap, AutoShot introduces the SHOT dataset specifically to capture short-video characteristics: brief duration, dense editing, vertical layouts, complex gradual transitions, and heavy special effects [4]. AutoShot leverages NAS [10], [15], [16] to search over approximately 83.9 million candidate architectures composed of 3-D separable convolution blocks and temporal Transformer blocks. While the searched backbone yields notable F1 improvements on SHOT, the computational cost of the search and retraining phases remains prohibitively high.

Imbalance, calibration, and temporal smoothing. Frame-level SBD is severely imbalanced, as boundary frames constitute a minuscule fraction of a video. Focal Loss has been widely adopted to reduce the contribution of easy, non-boundary samples, directing the learning focus toward hard positives and hard negatives [11]. Furthermore, probability calibration is crucial because SBD decisions rely directly on thresholding model scores. Temperature scaling offers a simple post-training calibration method that rescales logits without altering their ranking [12]. Finally, temporal smoothing effectively suppresses isolated noisy peaks, rendering boundary decisions more stable across neighboring frames in highly dynamic short videos.

III. The SBD Foundation: TransNetV2 and AutoShot

In this section, we first formulate the shot boundary detection (SBD) problem. Next, we detail the spatial-temporal factorized convolution, dilated convolutional blocks, and similarity features that form the backbone of modern SBD architectures. Finally, we review the Neural Architecture Search (NAS) strategy employed by AutoShot to select the optimal backbone.

A. Problem Formulation

Given a video represented as an ordered sequence of T frames $\{x_t\}_{t=1}^T$, SBD aims to predict a binary sequence $\{b_t\}_{t=1}^T$, where $b_t = 1$ denotes a shot boundary (either an abrupt cut or a gradual transition) at frame t, and $b_t = 0$ represents a non-boundary frame. An SBD model outputs a boundary probability sequence $\{p_t\}_{t=1}^T$ where $p_t \in [0, 1]$. The final boundary set is obtained by thresholding the probability scores and optionally grouping consecutive positive predictions:

$$\hat{b}_t = \begin{cases} 1, & p_t > \theta, \\ 0, & p_t \le \theta, \end{cases} \tag{1}$$

where the threshold θ is optimized on a validation set to balance precision and recall.

B. Spatial-Temporal Factorized Convolution

Deep learning models for SBD must process spatial-temporal representations directly from raw frame sequences. To avoid the high parameter count and computational complexity of standard 3-D convolutional kernels of size $k \times k \times k$, modern architectures factorize each 3-D convolution into two sequential operations:

1) A 2-D spatial convolution with kernel size $1 \times k \times k$. This processes each frame in the sequence independently, learning visual representation cues such as edges, textures, and color regions.
2) A 1-D temporal convolution with kernel size $k \times 1 \times 1$. This processes the spatial features along the temporal axis to capture motion patterns and scene changes.

With input/output channels C and kernel size $k = 3$, factorizing the kernel reduces the parameter count from $27C^2$ to $9C^2 + 3C^2 = 12C^2$. This represents a parameter reduction of approximately 56%, which mitigates overfitting and imposes an inductive bias that separates spatial contents from temporal motion.

C. Dilated DCNN Blocks

The backbone of the baseline architecture consists of stacked Dilated Convolutional Neural Network (DDCNN) blocks. Each block splits the input features into four parallel temporal convolution branches with dilation rates d increasing exponentially:

$$d \in \{1, 2, 4, 8\}. \tag{2}$$

The outputs of these four branches are concatenated along the channel dimension, followed by Batch Normalization (BatchNorm) and Rectified Linear Unit (ReLU) activation.

This multi-scale dilation design expands the temporal receptive field. For a temporal kernel size $K = 3$, the cumulative receptive field R_k after block k is calculated inductively as:

$$R_k = R_{k-1} + 2 \cdot (K - 1) \cdot \sum_{d \in \{1,2,4,8\}} d = R_{k-1} + 60, \tag{3}$$

with $R_0 = 1$ frame. Stacking six blocks theoretically yields a receptive field of up to 361 frames. In practice, videos are processed in sliding windows of 100 frames, so the effective temporal context is capped by the window length rather than the theoretical 361 frames. This context is wide enough to capture long gradual transitions, such as dissolves and fades.

D. Similarity Branch

To supplement the features learned by the convolutional blocks, a parallel similarity branch is integrated into the model. This branch extracts two forms of visual consistency:

1) RGB Histogram Similarity: A 512-bin color histogram is computed for each frame. The cosine distance between histograms of consecutive frames yields a 101-dimensional vector representing color shifts.
2) Learned Similarity: The cosine similarity between normalized deep features extracted from the intermediate layers of the convolutional blocks is calculated.

The histogram and learned similarities are fused and processed through a fully connected layer with 128 units and ReLU activation. The resulting similarity features are concatenated with the convolutional features before final classification.

E. AutoShot Backbone Search

While TransNetV2 employs a fixed configuration of identical DDCNN blocks, AutoShot [4] optimizes the backbone configuration for short-form videos using Neural Architecture Search (NAS). The search space covers four DDCNN block variants (differing in channel configurations and spatial feature sharing) across 6 layers, and a choice of 0 to 4 Transformer layers [20] at the 7th layer, resulting in 83.9×10^6 candidate architectures.

The search utilizes a Single-Path One-Shot supernetwork [22] combined with Bayesian Optimization. The baseline supernetwork is also compared against differentiable architectures like DARTS [23]. After searching, the model is further refined via knowledge distillation [19] and weight grafting. The optimal architecture found under the F1-score criterion (AutoShot@F1) consists of:

- Layer 1: DDCNNV2 (4 filters, dilation $nd = 4$).
- Layers 2–4: DDCNNV2A (4 filters, dilation $nd = 5$) shared spatial convolutions.
- Layer 5: DDCNNV2 (12 filters, dilation $nd = 5$).
- Layer 6: DDCNNV2 (8 filters, dilation $nd = 5$).
- Layer 7: 0 Transformer layers.

This backbone extracts a 4864-dimensional feature vector at the fc1_0 stage. In this paper, we freeze this optimized AutoShot backbone and denote the extracted features for frame t as:

$$h_t \in \mathbb{R}^{4864}. \tag{4}$$

IV. Proposed AutoShotV2: Calibrated SBD with Dual-Branch Head

AutoShotV2 improves the decision stage of SBD by training a new classification head and adding probability calibration and temporal smoothing post-processing steps. Fig. 1 illustrates the overall inference pipeline.

A. Network Architecture of the Classification Head

The proposed head takes the cached feature vector h_t and maps it to a shared hidden state:

$$u_t = \mathrm{ReLU}(W_f h_t + c_f), \tag{5}$$

where $W_f \in \mathbb{R}^{1024 \times 4864}$. A dropout layer with a keep probability of 0.5 is applied to u_t for regularization. Two parallel linear classification layers then output the logits:

$$z_t^{(1)} = W_1 u_t + c_1, \qquad z_t^{(2)} = W_2 u_t + c_2. \tag{6}$$

The primary branch ($z_t^{(1)}$) is trained with the exact one-hot boundary labels and is utilized during inference. The auxiliary branch ($z_t^{(2)}$) is supervised by boundary-aware many-hot targets to provide denser gradients. Fig. 2 shows the details of this head.

B. Focal Loss for Class Imbalance

Due to the extreme sparsity of boundary frames ($< 1\%$), standard Binary Cross-Entropy (BCE) loss tends to be dominated by easy non-boundary frames, causing the model to bias toward predicting negative classes. We address this class imbalance by optimizing the classification head using Focal Loss [11]:

$$\mathcal{L}_{\text{focal}} = -\alpha (1 - p_t^*)^\gamma \log(p_t^*), \tag{7}$$

where $p_t^* = p_t$ if the target $y_t = 1$ and $1 - p_t$ otherwise, with $p_t = \sigma(z_t)$. The dynamic scaling factor $(1 - p_t^*)^\gamma$ automatically down-weights the loss contributions of easy-to-classify frames and concentrates training gradients on hard boundaries and complex transitions. Grid search on the validation set yields the optimal parameters $\gamma = 1.0$ and $\alpha = 0.5$.

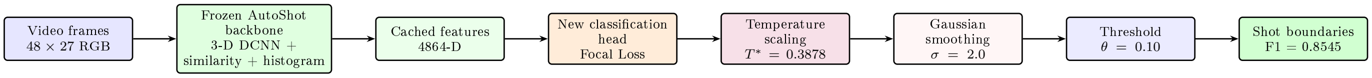

Fig. 1. AutoShotV2 inference pipeline. The searched AutoShot backbone is frozen. AutoShotV2 modifies the decision stage by training a new classification head and adding calibration and temporal smoothing before thresholding.

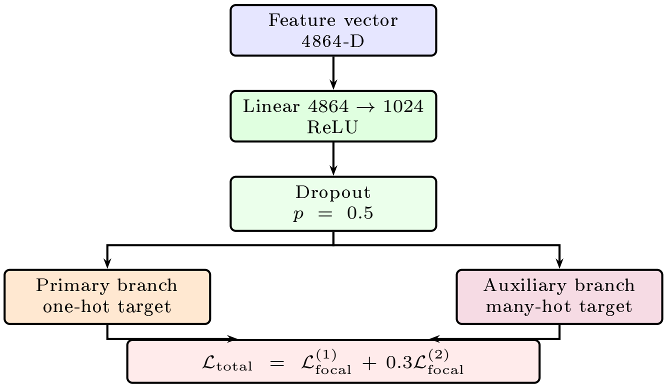

Fig. 2. AutoShotV2 classification head. The primary branch preserves sharp boundary localization, while the auxiliary branch supplies denser supervision around each boundary.

C. Boundary-Aware Many-Hot Supervision

One-hot annotations are temporally precise but sparse. They also do not account for labeling offsets or the transition span of gradual transitions. To address this, AutoShotV2 introduces an auxiliary many-hot supervision target $y_{\text{mh}}[t]$ that expands the positive labels to adjacent frames, inspired by label smoothing techniques [17]:

$$y_{\text{mh}}[t] = \begin{cases} 1, & \exists\, t_b \text{ s.t. } |t - t_b| \le 1, \\ 0, & \text{otherwise,} \end{cases} \tag{8}$$

where t_b represents a ground-truth boundary frame.

The overall training objective combines both branches:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{focal}}(\hat{y}^{(1)}, y_{\text{oh}}) + 0.3\,\mathcal{L}_{\text{focal}}(\hat{y}^{(2)}, y_{\text{mh}}). \tag{9}$$

The auxiliary weight 0.3 prevents the many-hot branch from over-smoothing the final boundary localization.

D. Probability Calibration and Post-Processing

Temperature Scaling. To calibrate the output confidence scores, the primary branch logit $z_t^{(1)}$ is scaled by a temperature parameter T:

$$p_t^{\text{cal}} = \sigma\left(\frac{z_t^{(1)}}{T}\right). \tag{10}$$

The optimal temperature T^* is found by minimizing the binary cross-entropy on the validation set:

$$T^* = \arg\min_{T\in[0.1,10.0]} \mathcal{L}_{\text{BCE}}\left(\sigma\left(\frac{z^{(1)}}{T}\right), y\right), \tag{11}$$

resulting in $T^* = 0.3878$. Because $T^* < 1$, temperature scaling sharpens the probability peaks for confident predictions.

Gaussian Temporal Smoothing. SBD models are prone to generating false-positive spikes due to single-frame disturbances (e.g., flash effects, camera shakes, quick occlusion). We apply a 1-D Gaussian filter to smooth the calibrated probability sequence:

$$\tilde{p}_t = \sum_k G_\sigma(k) p_{t-k}^{\text{cal}}, \tag{12}$$

where the Gaussian kernel is defined as:

$$G_\sigma(k) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{k^2}{2\sigma^2}\right). \tag{13}$$

We select $\sigma = 2.0$ via validation search, which suppresses isolated high-frequency false-alarm spikes while preserving temporally supported transition segments. EMA is not part of this deployment pipeline; it is evaluated separately in a full-model fine-tuning study and is therefore excluded from the controlled Phase 2 ablation.

V. Experiments

A. Implementation and Downstream Tasks

Datasets. The primary evaluation set is SHOT [4]. SHOT contains 853 complete short videos and 11,606 shot annotations, including 2,716 high-quality annotations in the 200-video test split. The dataset differs from traditional SBD benchmarks because the videos are short, vertically edited, fast-paced, and rich in visual effects. BBC Planet Earth and ClipShots are used to assess cross-domain behavior. BBC contains professional documentary footage with many smooth gradual transitions. ClipShots contains diverse open-world videos from YouTube and Weibo, with a longer average duration and many wild transition styles [8]. Table I summarizes the datasets used in this paper.

Metrics and implementation details. Performance is measured by precision, recall, and F1-score. F1-score is the main selection criterion because SBD involves a direct trade-off between false alarms and missed transitions. AutoShotV2 trains only the classification head using Adam optimizer with a learning rate of 10^{-5}, weight decay of 10^{-4}, and frame-level batch size of 512. Training runs for 20 epochs with a 4-epoch linear warm-up followed by cosine annealing [13]. The optimal threshold and temperature are calibrated using the validation set. Table II details the selected hyperparameters.

B. Main Results

Overall performance. Table III reports F1-scores across SHOT, BBC, and ClipShots. With the fixed deployment configuration ($T = 0.3878$, $\sigma = 2.0$, and $\theta = 0.10$), AutoShotV2 reaches 0.8545 on SHOT and 0.9656 on BBC.

TABLE I
Dataset Characteristics Used for SBD Evaluation

Property	BBC	ClipShots	SHOT
Content type	Documentary	Open-world online video	Short video
Total dataset videos	11	4039	853
Evaluation videos	11	500	200
Average video length	2945 s	237 s	39.5 s
Average shot length	6.57 s	15.34 s	2.59 s
Hard cuts	Many	128000+	8890
Gradual transitions	Many	38000+	2716
Main challenge	Gradual transitions	Diverse effects	Fast editing pace

TABLE II
AutoShotV2 Hyperparameters

Component	Value	Selection
Focal Loss γ	1.0	Grid search
Focal Loss α	0.5	Grid search
Learning rate	10^{-5}	Grid search
Weight decay	10^{-4}	Adam
Batch size	512	Frame-level
Maximum epochs	20	Fixed budget
Warm-up	4 epochs	Linear
Many-hot weight	0.3	Validation
Gaussian σ	2.0	Validation
Temperature T^*	0.3878	Validation BCE

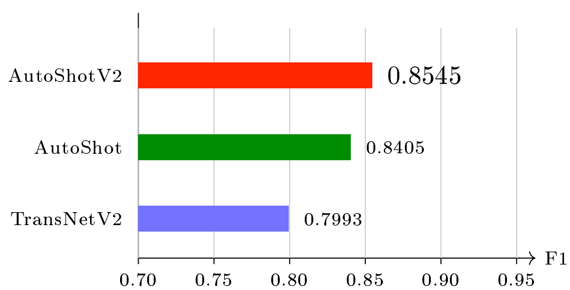

Fig. 3. F1-score comparison on SHOT. AutoShotV2 improves the reproduced AutoShot baseline by 1.40 points and the reported TransNetV2 baseline by 5.52 points.

On ClipShots, the fixed configuration reaches 0.7530; a separate test-set threshold sweep reaches 0.7557 at $\theta = 0.15$. The latter is reported for transparency and is not treated as the fixed deployment result. Reported baselines come from the original papers, whereas reproduced baselines use our evaluation pipeline. The TransNetV2 reproduction was not evaluated on SHOT.

Controlled ablation study. Table IV reports the Phase 2 ablation conducted on the cached-feature pipeline. A1 is the controlled BCE + one-hot reference. A0 is a historical AutoShot baseline evaluated with a per-dataset best-threshold sweep and should not be interpreted as a protocol-matched control. The B4 configuration is selected because it gives the strongest SHOT improvement while preserving the BBC guardrail. EMA is excluded because the available EMA experiment fine-tunes the full model on a different training pipeline.

C. Performance on Downstream Datasets

Effect on short-video data. The robust improvements on SHOT indicate that AutoShotV2 effectively models short-video characteristics. The Focal Loss tackles the extreme class imbalance between boundary and non-boundary frames, while the many-hot auxiliary target supplies denser gradients around ambiguous transitions without diluting precision. Simultaneously, Gaussian smoothing heavily suppresses isolated false peaks, which frequently occur in short videos due to rapid camera motion and strong visual editing effects.

Cross-domain behavior and ClipShots error analysis. While AutoShotV2 transfers well to the structured documentary footage of BBC, the fixed deployment configuration reaches only 0.7530 on the wild, open-world ClipShots dataset (Table III). To diagnose this, Table V splits ClipShots performance by transition type using the same fixed threshold $\theta = 0.10$.

The breakdown reveals that the primary source of degradation is the extremely low recall (4.42%) on gradual transitions. We attribute this to three potential causes: (1) the ± 1 many-hot window may be too narrow for ClipShots' extended gradual transitions; (2) the temperature learned on SHOT miscalibrates the logit distribution for ClipShots; and (3) the chosen Focal Loss parameters heavily emphasize SHOT's specific hard examples over open-world editing styles.

VI. Conclusion and Discussion

In this paper, we presented AutoShotV2, a compact refinement framework for short-video shot boundary detection. By freezing the AutoShot backbone and optimizing a two-branch classification head with Focal Loss and many-hot supervision, the method addresses the extreme class imbalance characteristic of short videos. Temperature scaling and Gaussian smoothing provide the calibrated decision pipeline used at deployment. AutoShotV2 raises SHOT F1 from 0.8405 for the reproduced AutoShot baseline to 0.8545, the best result among the evaluated methods, without repeating the computationally expensive neural architecture search.

TABLE III
F1-score Comparison Across Evaluation Datasets

Method	SHOT	BBC	ClipShots
AutoShot, reported in original paper	0.8410	0.9710	0.7870
AutoShot, reproduced	0.8405	0.9554	0.7753
TransNetV2, reported baseline	0.7993	0.9620	0.7760
TransNetV2, reproduced	–	0.9593	0.7454
AutoShot reproduced + Gaussian smoothing	0.8480	0.9510	0.7987
AutoShotV2, fixed deploy threshold	0.8545	0.9656	0.7530
AutoShotV2, ClipShots best sweep	–	–	0.7557

TABLE IV
Controlled Phase 2 Ablation (Separate from the Deploy Checkpoint)

Configuration	SHOT	BBC	ClipShots
A0 – AutoShot baseline	0.8405	0.9554	0.7649
A1 – BCE + one-hot	0.8378	0.9570	0.6983
A2 – Focal only	0.8378	0.9567	0.6967
A3 – Many-hot only	0.8375	0.9563	0.7005
P1 – Gaussian only	0.8432	0.9436	0.7519
P2 – Temperature only	0.8378	0.9570	0.6983
B1 – Focal + many-hot	0.8384	0.9559	0.7006
B4 – Temperature + Gaussian (selected)	0.8540	0.9570	0.7441
B5 – Full candidate, no EMA	0.8542	0.9551	0.7409

TABLE V
ClipShots Breakdown by Transition Type at the Fixed Deploy Threshold

Transition	Prec.	Rec.	F1
Cut	0.5826	0.8860	0.7030
Gradual	0.6456	0.0442	0.0828

Despite these advancements, several challenges remain. Cross-domain analysis revealed a noticeable performance degradation on diverse open-world datasets like ClipShots, primarily due to missed gradual transitions. This emphasizes that SBD systems should treat calibration, loss balancing, and temporal windows as domain-sensitive parameters rather than fixed constants. Fast camera motion and game-like visual effects can still trigger false positives, indicating a limit to purely visual features. In the future, we aim to explore domain-adaptive calibration techniques and multimodal SBD architectures that integrate audio cues to further improve cross-domain robustness.

Acknowledgment

The authors thank the Faculty of Information Technology, University of Science, Vietnam National University Ho Chi Minh City, for supporting this research.

S. M. Tang and T. M. Do are with the Faculty of Information Technology, University of Science, Vietnam National University Ho Chi Minh City, Vietnam.
T. S. Tran supervised this research.

Manuscript prepared June 7, 2026.

References

[1] H. Seridi, M. Kurulay, Z. Kouahla, B. Farou, M. N. Kouahla, M. A. Ferrag, and M. A. Boudia, "Shot boundary detection: Fundamental concepts and survey," in CEUR Workshop Proceedings, vol. 2589, 2020.
[2] T. Kar, P. Kanungo, S. N. Mohanty, S. Groppe, and J. Groppe, "Video shot-boundary detection: Issues, challenges and solutions," Artificial Intelligence Review, vol. 57, no. 104, 2024.
[3] S. H. Abdulhussain, "Methods and challenges in shot boundary detection," Entropy, vol. 20, no. 4, p. 214, 2018.
[4] W. Zhu, Y. Yu, W. Huang, J. Huang, W. Feng, X. Wang, X. Yang, C. Hu, C. Liu, B. Fan, and T. He, "AutoShot: A short video dataset and state-of-the-art shot boundary detection," in Proc. IEEE/CVF CVPR Workshops, 2023.
[5] T. Soucek and J. Moravec, "TransNet V2: An effective deep network architecture for fast shot transition detection," arXiv preprint arXiv:2008.04838, 2020.
[6] T. Soucek, J. Moravec, and J. Lokoc, "TransNet: A deep network for fast detection of common shot transitions," arXiv preprint arXiv:1906.03363, 2019.
[7] L. Baraldi, C. Grana, and R. Cucchiara, "Shot and scene detection via hierarchical clustering for re-using broadcast video," in Computer Analysis of Images and Patterns, 2015, pp. 801–811.
[8] S. Tang, L. Feng, Z. Kuang, Y. Chen, and W. Zhang, "Fast video shot transition localization with deep structured models," arXiv preprint arXiv:1808.04234, 2018.
[9] A. Hassanien, M. Elgharib, A. Selim, S.-H. Bae, M. Hefeeda, and W. Matusik, "Large-scale, fast and accurate shot boundary detection through spatio-temporal convolutional neural networks," arXiv preprint arXiv:1705.03281, 2017.
[10] T. Elsken, J. H. Metzen, and F. Hutter, "Neural architecture search: A survey," Journal of Machine Learning Research, vol. 20, no. 55, pp. 1–21, 2019.
[11] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," in Proc. IEEE International Conference on Computer Vision, 2017, pp. 2980–2988.
[12] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in Proc. International Conference on Machine Learning, 2017, pp. 1321–1330.

[13] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in Proc. International Conference on Learning Representations, 2017.
[14] V. Anh, "Short-form video trend harming youth," VietnamNet Global, May 2023.
[15] C. White, M. Safari, R. Sukthanker, B. Ru, T. Elsken, A. Zela, D. Dey, and F. Hutter, "Neural architecture search: Insights from 1000 papers," arXiv preprint arXiv:2301.08727, 2023.
[16] P. Ren, Y. Xiao, X. Chang, P.-Y. Huang, Z. Li, X. Chen, and X. Wang, "A comprehensive survey of neural architecture search: Challenges and solutions," arXiv preprint arXiv:2006.02903, 2020.
[17] R. Müller, S. Kornblith, and G. E. Hinton, "When does label smoothing help?" in Proc. Advances in Neural Information Processing Systems, 2019, pp. 4694–4703.
[18] A. Esteve et al., "Shot boundary detection with 3D depthwise convolutions and visual attention," Sensors, vol. 23, no. 12, p. 5620, 2023.
[19] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," arXiv preprint arXiv:1503.02531, 2015.
[20] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in Proc. Advances in Neural Information Processing Systems, 2017, pp. 5998–6008.
[21] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri, "Learning spatiotemporal features with 3D convolutional networks," in Proc. IEEE International Conference on Computer Vision, 2015, pp. 4489–4497.
[22] Z. Guo, X. Zhang, H. Mu, W. Heng, Z. Liu, Y. Wei, and J. Sun, "Single path one-shot neural architecture search with uniform sampling," in Proc. European Conference on Computer Vision, 2020, pp. 544–560.
[23] H. Liu, K. Simonyan, and Y. Yiming, "DARTS: Differentiable architecture search," in Proc. International Conference on Learning Representations, 2019.